EXPERIMENT 17

CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB

Linear Regression

Aim: Predict salary based on years of experience using Linear Regression.

Python Program

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
df=pd.read_csv('Salary_data.csv').dropna()
x=df[['YearsExperience']]
y=df['Salary']
xtr,xte,ytr,yte=train_test_split(x,y,test_size=0.2,random_state=1)
model=LinearRegression()
model.fit(xtr,ytr)
print("Accuracy:",model.score(xte,yte))
print("Prediction:",model.predict([[5]]))
```

Result

The model is trained and predicts salary for a selected experience value.

Note: For CSV programs, keep the named CSV file in the same folder and check that column names match.